INMO(31^{st}) – 2016

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Let ABC be a triangle in which $AB = AC$. Suppose the orthocenter of the triangle lies on the incircle. Find the ratio AB/BC.
2. For positive real numbers a,b,c which of the following statement implies $a=b=c$. Justify your answer.
   (a) $a(b^3+c^3)=b(c^3+a^3)=c(a^3+b^3)$.
   (b) $a(a^3+b^3)=b(b^3+c^3)=c(c^3+a^3)$.
3. Let N denotes the set of all natural numbers. Define a function $T: N \to N$ by $T(2k)=k$ and $T(2k+1)=2k+2$. We write $T^2(n)=T(T(n))$ and in general $T^k(n)=T^{k-1}(T(n))$ for any $k>1$.
   (a) Show that for each $n \in N$, there exist k such that $T^k(n)=1$.
   (b) For $k \in N$, let c_k denote the number of elements in the set $a(\{n : T^k(n)=1\}$.
   Prove that $c_{k+2}=c_{k+1}+c_k$, for $k \geq 1$.
4. Suppose 2016 points of the circumference of a circle are colored red and the remaining points are colored blue. Given any natural number $n \geq 3$, prove that there is a regular n-sided polygon all of whose vertices are blue.
5. Let ABC be a right angled triangle with $\angle B = 90^0$. Let D be a point on AC such that the inradii of the triangle ABD and CBD are equal. If this common value is r' and r is the inradii of triangle ABC, prove that $\frac{1}{r'}=\frac{1}{r}+\frac{1}{BD}$.
6. Consider a non constant arithmetic progression $a_1, a_2, ..., a_n, ...$ Suppose there exist relatively prime positive integer $p>1$ and $q>1$ such that $a_1^{\,2}, a_{p+1}^{\,2}, a_{q+1}^{\,2}$ are also the terms of the same arithmetic progression. Prove that the terms of the arithmetic progression are all integers.

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